

COMPETITIVE EXAMINATION PREPARATION
BPS-17 Lecturer Mathematics

FPSC PPSC AJKPSC KPSC
Federal Punjab AJK KPK

MATHEMATICS
CORE SYLLABUS

Official Examination Framework 2022

PAPER – I

Section A:

Calculus & Differential Equations

Section B:

Real Analysis, Complex Analysis
& Calculus II (Plane Geometry)

PAPER – II

Section A:

Group Theory, Ring Theory,
Linear Algebra & Vector Spaces

Section B:

Metric Spaces, Functional Analysis,
Linear Algebra & Numerical Analysis

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Reference Material — PSC 2022–26

**Table of Contents****Contents**

1	Section A: Calculus & Differential Equations	2
1.1	Differential Calculus	2
1.2	Integral Calculus	3
1.3	Ordinary Differential Equations (ODEs)	3
2	Section B: Real Analysis, Complex Analysis & Calculus II	5
2.1	Sequences, Series & Power Series	5
2.2	Complex Analysis	5
2.3	Calculus II: Plane Geometry & Curves	6
3	Section A: Group Theory, Ring Theory, Linear Algebra & Vector Spaces	7
3.1	Group Theory	7
3.2	Rings, Fields & Integral Domains	8
3.3	Vector Spaces & Linear Independence	8
4	Section B: Metric Spaces, Functional Analysis, Linear Algebra & Numerical Analysis	9
4.1	Metric Spaces & Topological Spaces	9
4.2	Normed Spaces, Inner Product Spaces & Gram–Schmidt Process	9
4.3	Matrices & Linear Algebra	10
4.4	Numerical Analysis	10



PAPER – I

Applied and Analytical Mathematics

Section A: Calculus & Differential Equations

Differential Calculus

Limits & Continuity

- Definition and properties of **Limits**; one-sided limits
- **Continuity** of functions: definition, types of discontinuity
- Continuity on closed intervals; uniform continuity

Differentiability & Applications

- **Differentiability**: definition, relationship with continuity
- Rules of differentiation: product, quotient, chain rules
- Implicit and parametric differentiation
- Higher-order derivatives

General Theorems

- **Rolle's Theorem**: statement, conditions, and geometric interpretation
- **Mean Value Theorem** (Lagrange's): statement and applications
- Cauchy's Mean Value Theorem
- Taylor's Theorem with Maclaurin's Series

Asymptotes, Maxima & Minima

- Horizontal, vertical, and oblique **asymptotes**
- **Maxima and Minima**: first and second derivative tests
- Applied optimization problems
- Concavity, inflection points, curve sketching

Integral Calculus

Definite & Indefinite Integrals

- **Indefinite integrals:** standard forms, substitution, partial fractions
- **Definite integrals:** properties, fundamental theorem of calculus
- Integration by parts; reduction formulae
- Improper integrals: types and convergence

Applications of Integration

- **Quadrature:** areas under curves (Cartesian & polar)
- **Rectification:** arc length of curves
- **Areas and Volumes:** volumes of revolution (disk, shell, washer methods)
- **Centre of Mass:** centroid calculations using integration

Numerical Integration

- **Trapezoidal Rule:** formula, error estimation
- **Simpson's $\frac{1}{3}$ Rule** and **Simpson's $\frac{3}{8}$ Rule:** derivation and applications
- Comparison of accuracy; composite rules

Multiple Integrals & Riemann–Stieltjes Integral

- **Double integrals:** Cartesian and polar forms; change of order
- **Triple integrals:** cylindrical and spherical coordinates
- Applications: volumes, mass, moments of inertia
- **Riemann–Stieltjes Integral:** definition, properties, and existence conditions

Ordinary Differential Equations (ODEs)

First-Order ODEs & Applications

- Separable, linear, exact, homogeneous, and Bernoulli equations
- Integrating factors; solution methods
- **Rectilinear Motion:** Newton's second law, velocity, acceleration problems
- **Growth & Decay Problems:** population dynamics, radioactive decay, Newton's law of cooling

Second-Order ODEs & Applications

- Homogeneous and non-homogeneous second-order linear ODEs
- Methods: undetermined coefficients, variation of parameters
- **Spring–Mass System:** free and forced vibrations, damping
- **Simple Harmonic Oscillator:** equation of motion, general solution, amplitude and phase
- Resonance and steady-state solutions

Key Formulae — Section A**Rolle's Theorem:**

$f(a) = f(b)$, f cont. on $[a, b]$,
diff. on $(a, b) \Rightarrow \exists c \in (a, b): f'(c) = 0$

MVT: $f'(c) = \frac{f(b) - f(a)}{b - a}$

Trapezoidal Rule:

$$\int_a^b f \, dx \approx \frac{h}{2} [f_0 + 2f_1 + \cdots + 2f_{n-1} + f_n]$$

Simpson's $\frac{1}{3}$ Rule:

$$\int_a^b f \, dx \approx \frac{h}{3} [f_0 + 4f_1 + 2f_2 + \cdots + f_n]$$

Simple Harmonic Oscillator:

$$\ddot{x} + \omega^2 x = 0, \quad x(t) = A \cos(\omega t + \phi)$$

Spring–Mass:

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

Growth/Decay:

$$\frac{dN}{dt} = kN \Rightarrow N(t) = N_0 e^{kt}$$

Riemann–Stieltjes:

$$\int_a^b f \, d\alpha = \lim_{\|\mathcal{P}\| \rightarrow 0} \sum f(\xi_i) \Delta \alpha_i$$



Section B: Real Analysis, Complex Analysis & Calculus II

Sequences, Series & Power Series

Sequences & Series

- Definition of sequence; bounded, monotone, convergent sequences
- Cauchy criterion for convergence
- Infinite series: partial sums, convergence and divergence

Convergence Tests

- **Comparison Test, Limit Comparison Test**
- **Ratio Test** (D'Alembert) and **Root Test** (Cauchy)
- **Integral Test, Alternating Series Test** (Leibniz)
- Absolute vs. conditional convergence

Power Series

- Definition of power series $\sum_{n=0}^{\infty} a_n(x - c)^n$
- **Radius of Convergence:** Cauchy-Hadamard formula $R = \frac{1}{\limsup |a_n|^{1/n}}$
- **Interval of Convergence:** endpoint analysis
- Differentiation and integration of power series; Taylor and Maclaurin series

Complex Analysis

Functions of a Complex Variable

- Complex numbers: algebra, modulus, argument, polar form
- **De Moivre's Theorem:** $(re^{i\theta})^n = r^n e^{in\theta}$; applications to roots of unity
- Complex functions; limits and continuity in $\mathbb{C}$

Analytic Functions & Singularities

- **Analytic functions:** Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$
- Harmonic functions; harmonic conjugates
- **Singularities:** removable, poles, essential singularities
- Laurent series expansion; residues

Complex Integration

- Contour integrals; path independence
- **Cauchy's Theorem:** $\oint_C f(z) dz = 0$ for analytic f
- **Cauchy's Integral Formula:** $f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z - z_0} dz$
- Residue theorem; evaluation of real integrals using residues

Calculus II: Plane Geometry & Curves**Conic Sections**

- **Cartesian coordinates:** standard forms of circle, ellipse, parabola, hyperbola
- **Plane Polar Coordinates:** relation to Cartesian; polar equations of conics
- General second-degree equation; rotation and translation of axes

Space Curves & Differential Geometry

- **Vector equations** for plane and space curves: $\mathbf{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$
- **Tangent, Normal, and Binormal** vectors (TNB frame)
- **Curvature κ and Torsion τ :** $\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$, $\tau = \frac{(\mathbf{r}' \times \mathbf{r}'') \cdot \mathbf{r}'''}{|\mathbf{r}' \times \mathbf{r}''|^2}$
- **Serret–Frenet Formulae:** $\mathbf{T}' = \kappa\mathbf{N}$, $\mathbf{N}' = -\kappa\mathbf{T} + \tau\mathbf{B}$, $\mathbf{B}' = -\tau\mathbf{N}$

PAPER – II

Pure Mathematics & Computational Methods

Section A: Group Theory, Ring Theory, Linear Algebra & Vector Spaces

Group Theory

Foundations of Groups

- **Definition of a Group:** closure, associativity, identity, inverse
- **Examples:** $(\mathbb{Z}, +)$, $(\mathbb{Q}^*, \times)$, S_n , $GL(n, \mathbb{R})$, dihedral groups D_n
- **Order of a Group $|G|$; Order of an Element $o(a)$**
- **Abelian (Commutative) Groups vs. Non-Abelian Groups**
- **Cyclic Groups:** definition, generators, $\mathbb{Z}_n$; every subgroup of a cyclic group is cyclic

Subgroups & Cosets

- Subgroup criteria; Lagrange's theorem: $|H|$ divides $|G|$
- Cosets: left and right cosets; index of a subgroup
- **Normal Subgroups:** $N \trianglelefteq G$ iff $gNg^{-1} = N \forall g \in G$
- **Characteristic Subgroups:** invariant under all automorphisms
- **Normalizer $N_G(H)$ and Centralizer $C_G(a)$ in a group**

Homomorphisms & Isomorphisms

- **Homomorphism:** $\phi(ab) = \phi(a)\phi(b)$; kernel and image
- **Fundamental Theorem of Homomorphism:** $G/\ker \phi \cong \phi(G)$
- **First, Second & Third Isomorphism Theorems** for groups
- **Automorphisms:** inner automorphisms; automorphism group $\text{Aut}(G)$



Rings, Fields & Integral Domains

Rings & Subrings

- **Definition of a Ring:** two binary operations; ring axioms
- **Examples:** $(\mathbb{Z}, +, \times)$, $\mathbb{Z}_n$, matrix rings $M_n(\mathbb{R})$, polynomial rings $\mathbb{R}[x]$
- **Subrings:** subring test; ideals and quotient rings
- **Integral Domains:** commutative ring with no zero divisors

Fields

- **Field:** commutative division ring; examples $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Z}_p$ (p prime)
- **Field Extensions:** algebraic and transcendental extensions
- **Finite Fields (Galois Fields) $GF(p^n)$:** examples $GF(2)$, $GF(4)$
- **Characteristic** of a field; prime subfields

Vector Spaces & Linear Independence

- **Vector Spaces:** axioms, examples over $\mathbb{R}$ and $\mathbb{C}$
- **Linear Independence and Dependence:** definition, Wronskian test
- **Basis and Dimension:** spanning sets, basis theorem, coordinates
- Examples of **finite and infinite dimensional vector spaces**
- **Subspaces:** sum and intersection; direct sums



Section B: Metric Spaces, Functional Analysis, Linear Algebra & Numerical Analysis

Metric Spaces & Topological Spaces

Metric Spaces

- **Definition:** (X, d) with non-negativity, symmetry, triangle inequality
- **Examples:** Euclidean $\mathbb{R}^n$, discrete metric, function spaces $C[a, b]$
- **Open and Closed Spheres:** $B(x, r)$, $\bar{B}(x, r)$
- **Interior, Exterior, Frontier (Boundary)** of a set
- **Sequences in Metric Spaces:** convergence, Cauchy sequences
- Completeness; complete metric spaces; Banach fixed-point theorem

Topological Spaces

- **Definition:** collection τ satisfying open-set axioms
- **Examples:** discrete, indiscrete, cofinite topologies
- Open and closed sets; closure, interior, boundary of a set
- Basis for a topology; subspace topology
- **Convergence of Sequences** in topological spaces; Hausdorff spaces

Normed Spaces, Inner Product Spaces & Gram–Schmidt Process

Normed & Banach Spaces

- **Normed Space:** $(X, \|\cdot\|)$; norm axioms, examples
- Every normed space is a metric space via $d(x, y) = \|x - y\|$
- **Banach Space:** complete normed space; examples ℓ^p , L^p , $C[a, b]$

Inner Product Spaces & Hilbert Spaces

- **Inner Product:** $\langle \cdot, \cdot \rangle$; Cauchy–Schwarz inequality $|\langle u, v \rangle| \leq \|u\| \|v\|$
- Orthogonality, orthonormal sets, orthonormal basis



- **Hilbert Space:** complete inner product space; $L^2[a, b]$
- **Gram–Schmidt Orthonormalization Process:** algorithm and examples

$$\mathbf{e}_k = \frac{\mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{e}_j \rangle \mathbf{e}_j}{\left\| \mathbf{v}_k - \sum_{j=1}^{k-1} \langle \mathbf{v}_k, \mathbf{e}_j \rangle \mathbf{e}_j \right\|}$$

Matrices & Linear Algebra

Linear Transformations & Matrices

- **Linear Transformations:** definition, kernel, range, rank-nullity theorem
- **Matrix representation** of linear transformations; change of basis
- **Matrix Algebra:** addition, multiplication, transpose, inverse
- **Row Reduction:** Echelon Form (EF) and Reduced Echelon Form (REF)

Systems of Linear Equations

- **Homogeneous Systems:** $AX = \mathbf{0}$; non-trivial solutions, null space
- **Non-Homogeneous Systems:** $AX = B$; existence and uniqueness (Rouché–Capelli)
- Gaussian elimination; back substitution

Determinants, Eigenvalues & Diagonalization

- **Properties of Determinants:** cofactor expansion, row operations
- **Eigenvalues and Eigenvectors:** characteristic equation $\det(A - \lambda I) = 0$
- **Diagonalization of Symmetric Matrices:** spectral theorem, orthogonal diagonalization
- Applications: quadratic forms, principal axes

Numerical Analysis

Numerical Methods for Linear Systems

- **Gauss–Seidel Method:** iterative solver for $AX = B$; convergence conditions



(diagonal dominance)

- **Jacobi Method:** simultaneous iteration; comparison with Gauss–Seidel
- Convergence criteria; spectral radius condition $\rho(B) < 1$
- Error analysis; stopping criteria

Syllabus at a Glance

Paper	Topics
Paper I – Sec A	Limits, Continuity, Differentiability, Rolle’s & MVT, Asymptotes, Maxima/Minima, Definite/Indefinite Integrals, Quadrature, Rectification, Trapezoidal & Simpson’s Rules, Multiple Integrals, Areas & Volumes, Centre of Mass, Riemann–Stieltjes Integral, 1st & 2nd Order ODEs, Rectilinear Motion, Growth/Decay, Spring–Mass & SHO
Paper I – Sec B	Sequences & Series, Convergence Tests, Power Series, Radius/Interval of Convergence, Complex Variables, De Moivre’s Theorem, Analytic Functions, Singularities, Cauchy’s Theorem, Cauchy’s Integral Formula, Conics (Cartesian & Polar), Space Curves, TNB Frame, Curvature, Torsion, Serret–Frenet Formulae
Paper II – Sec A	Groups (definition, order, cyclic, Abelian/non-Abelian), Lagrange’s Theorem, Normal Subgroups, Characteristic Subgroups, Normalizer, Centralizer, Fundamental Theorem of Homomorphism, Isomorphism Theorems, Automorphisms, Rings, Subrings, Integral Domains, Fields, Vector Spaces, Linear Independence, Basis & Dimension, Field Extensions, Finite Fields
Paper II – Sec B	Metric & Topological Spaces, Open/Closed Spheres, Interior/Exterior/Frontier, Sequence Convergence, Normed Spaces, Inner Product Spaces, Gram–Schmidt Process, Linear Transformations, Matrices, Echelon Forms, Homogeneous & Non-Homogeneous Systems, Determinants, Eigenvalues & Eigenvectors, Diagonalization of Symmetric Matrices, Gauss–Seidel Method, Jacobi Method

Mathematics

The Language of the Universe

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“Pure mathematics is, in its way, the poetry of logical ideas.”

— *Albert Einstein*